Peter Hagen

Annandale, VA | 703-332-9441 | U.S. citizen | Open to DMV hybrid/onsite and remote
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Professional Summary

Systems-engineering-trained analyst with 5+ years turning ambiguous operational questions into documented metrics, validated data workflows, process improvements, and decision-ready reporting. Brings SQL, Python, Tableau, Power BI, data quality, stakeholder communication, and federal procurement-adjacent analysis. Experienced documenting report logic, assumptions, and handoffs for technical and nontechnical audiences; U.S. citizen and clearance-eligible.

Education

George Mason University **Fairfax, VA**
M.S. Systems Engineering, GPA 3.98 *2019*

Virginia Tech **Blacksburg, VA**
B.S. Industrial and Systems Engineering, Minor in Business *2015*

Work Experience

Independent Analytics and Data Engineering Projects **Annandale, VA**
Builder, Portfolio and Consulting Projects *Aug 2025 – Present*

- Designed and shipped Recompete Radar (Sunlight), a federal-contracting analytics platform (Snowflake, dbt-core, Apache Airflow, Streamlit in Snowflake, Python, Postgres/Neon, FastAPI, Next.js 14/TypeScript) that models 484,242 prime award records and SAM.gov enrichment data across 20 civilian agencies and 12 service categories, ranking 27,673 expiring contracts with backtested scorecards.
- Built a transparent scoring model and self-auditing SQL analysis layer that tests assumptions against backtested data, producing traceable decision support rather than black-box output.

National Association of Home Builders **Washington, DC**
Senior Data Analyst *Jul 2024 – Jul 2025*

- Built Tableau and Power BI dashboards used by executive leadership to monitor retention, growth, regional performance, and operational priorities across 500+ local associations.
- Wrote SQL queries and validation logic across 114,000+ membership records for recurring reporting, ad hoc analysis, cleanup, deduplication, and source-of-truth metric definitions.
- Translated ambiguous leadership questions into dashboard requirements, data-review steps, reporting logic, visual summaries, and plain-language recommendations.
- Led data-quality initiatives across legacy membership systems, documenting assumptions and improving trust in recurring reports.

National Association of Home Builders **Washington, DC**
Director, Membership Data *Feb 2023 – Jul 2024*

- Managed reporting operations across 114,000+ member records, serving as the primary point of contact for membership data questions, reporting logic, and stakeholder data requests.
- Partnered with leadership, marketing, and membership teams to define KPIs and align analytics with strategic goals.
- Automated recurring reporting workflows with Python and SQL, reducing manual preparation while improving consistency and reliability.

Interim Work and Professional Development **Annandale, VA**
Seasonal E-Commerce Operations; Self-Directed Training *Nov 2020 – Feb 2023*

- Supported family members through the COVID-19 pandemic while completing self-directed training in Python, SQL, and cloud architecture to move from business analysis toward data and analytics work.
- Seasonal e-commerce operations at Olio2Go, a Merrifield, VA specialty food importer: fulfilled customer orders across a Shopify storefront and an Amazon marketplace channel through peak holiday periods.
- Delivery driver, Amazon.

Defense Government Contracting International **McLean, VA**
Business Analyst *Apr 2019 – Nov 2020*

- Built Python and Excel models to evaluate DoD fuel-bid outcomes, procurement scenarios, pricing assumptions, win probabilities, and logistics tradeoffs.
- Prepared decision-support materials for executives and procurement stakeholders, translating quantitative findings into recommended actions, open questions, and operational constraints.

Jovie **Fairfax, VA**
Business Analyst *Nov 2016 – Jul 2018*

- Built SQL analytics and Python/R ETL scripts for workforce allocation, scheduling coverage, service delivery, and operational reporting across 200+ employees.
- Created reporting and analysis tools that improved visibility into staffing workflows, resource gaps, and service efficiency.

Technical Skills

Analysis and systems: SQL, Python, R, Excel, Tableau, Power BI, KPI definition, dashboard design, data quality, reporting automation, data modeling

Process and documentation: reporting logic, metric definitions, workflow analysis, data validation, cleanup notes, stakeholder-ready findings, handoff guidance

Data and development: Snowflake, PostgreSQL, dbt-core, Apache Airflow, Streamlit, ETL/ELT concepts, REST APIs, AWS (S3, Lambda, EC2), Git, CI/CD, Docker, FastAPI, Next.js 14/TypeScript

Context: federal procurement analysis, executive reporting, cross-functional stakeholder collaboration, U.S. citizen and clearance-eligible